NAME: VINAY BHARGAV REDDY ID NO:2400031280 SEC NO : 12

FULL STACK DEVLOPMENT

Skill-3: Hibernate HQL Demo (Students, MySQL)

Purpose
Demonstrate Hibernate with HQL: seed products, paginate, count, group, min/max, and range queries.

Tech stack
Java 21, Maven, Hibernate 6.4.x, MySQL Connector/J, SLF4J Simple.

Project files (full contents)

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://www.w3.org/2001/XMLSchema-instance">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.demo</groupId>
  <artifactId>skill-3-hql</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <maven.compiler.release>21</maven.compiler.release>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <hibernate.version>6.4.8.Final</hibernate.version>
    <mysql.connector.version>8.0.33</mysql.connector.version>
    <jakarta.persistence.version>3.1.0</jakarta.persistence.version>
  </properties>

  <dependencies>
    <dependency>
      <groupId>org.hibernate.orm</groupId>
      <artifactId>hibernate-core</artifactId>
      <version>${hibernate.version}</version>
    </dependency>
    <dependency>
```

2400031280

```
        <groupId>com.mysql</groupId>
        <artifactId>mysql-connector-j</artifactId>
        <version>${mysql.connector.version}</version>
    </dependency>
    <dependency>
        <groupId>jakarta.persistence</groupId>
        <artifactId>jakarta.persistence-api</artifactId>
        <version>${jakarta.persistence.version}</version>
    </dependency>
    <dependency>
        <groupId>org.slf4j</groupId>
        <artifactId>slf4j-simple</artifactId>
        <version>2.0.16</version>
    </dependency>
</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.13.0</version>
            <configuration>
                <release>${maven.compiler.release}</release>
            </configuration>
        </plugin>
        <plugin>
            <groupId>org.codehaus.mojo</groupId>
            <artifactId>exec-maven-plugin</artifactId>
            <version>3.5.0</version>
            <configuration>
                <mainClass>com.demo.ProductHqlDemo</mainClass>
            </configuration>
        </plugin>
```

240031280

```
        </plugins>
    </build>
</project>
```

Student.java

```
package com.klef;

import jakarta.persistence.*;

@Entity
@Table(name="student")
public class Student {

@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)
private Long id;

private String name;
private String department;
private Double marks;
private Integer age;

public Student(){}

public Student(String name,String department,Double marks,Integer age){
this.name=name;
this.department=department;
this.marks=marks;
this.age=age;
}

public Long getId(){ return id; }

public String getName(){ return name; }
```

2400031280

```
public void setName(String name){ this.name=name; }

public String getDepartment(){ return department; }
public void setDepartment(String department){ this.department=department; }

public Double getMarks(){ return marks; }
public void setMarks(Double marks){ this.marks=marks; }

public Integer getAge(){ return age; }
public void setAge(Integer age){ this.age=age; }

@Override
public String toString(){
return id+" | "+name+" | "+department+" | "+marks+" | "+age;
}

}
```

StudentHqlDemo.java

```
package com.klef;

import java.util.List;
import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.Transaction;
import org.hibernate.cfg.Configuration;

public class StudentHqlDemo {

private static final SessionFactory factory=new Configuration()
.configure("hibernate.cfg.xml")
.addAnnotatedClass(Student.class)
.buildSessionFactory();
```

240031280

```
public static void main(String[] args){

seedData();
runQueries();

factory.close();
}

private static void seedData(){

List<Student> students=List.of(
new Student("Vinay","CSE",85.5,20),
new Student("Rahul","ECE",78.0,21),
new Student("Ankit","AI",90.2,22),
new Student("Divya","CSE",88.0,21),
new Student("Kiran","ECE",72.5,23),
new Student("Sneha","AI",95.0,20)
);

Session session=factory.openSession();
Transaction tx=session.beginTransaction();

session.createMutationQuery("delete from Student").executeUpdate();

students.forEach(session::persist);

tx.commit();
session.close();

System.out.println("Students Inserted\n");
}

private static void runQueries(){
```

2400031280

```
Session session=factory.openSession();

System.out.println("--- First 3 Students ---");

session.createQuery("from Student s order by s.id",Student.class)
.setFirstResult(0)
.setMaxResults(3)
.list()
.forEach(System.out::println);

System.out.println("\n--- Next 3 Students ---");

session.createQuery("from Student s order by s.id",Student.class)
.setFirstResult(3)
.setMaxResults(3)
.list()
.forEach(System.out::println);

Long total=session.createQuery("select count(s) from Student s",Long.class).uniqueResult();

System.out.println("\nTotal Students: "+total);

System.out.println("\n--- Group By Department ---");

session.createQuery(
"select s.department,count(s) from Student s group by s.department",
Object[].class)
.list()
.forEach(r->System.out.println(r[0]+" -> "+r[1]));

Object[] minmax=session.createQuery(
"select min(s.marks),max(s.marks) from Student s",
Object[].class)
.uniqueResult();
```

2400031280

```
System.out.println("\nMin Marks: "+minmax[0]);
System.out.println("Max Marks: "+minmax[1]);

System.out.println("\n--- Marks Between 80 and 95 ---");

session.createQuery(
"from Student s where s.marks between :min and :max",
Student.class)
.setParameter("min",80.0)
.setParameter("max",95.0)
.list()
.forEach(System.out::println);

session.close();

}

}
```

hibernate.cfg.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE hibernate-configuration PUBLIC
    "-//Hibernate/Hibernate Configuration DTD 3.0//EN"
    "http://www.hibernate.org/dtd/hibernate-configuration-3.0.dtd">
<hibernate-configuration>
  <session-factory>
    <property name="connection.driver_class">com.mysql.cj.jdbc.Driver</property>
    <property
name="connection.url">jdbc:mysql://localhost:3306/hibernatedb?useSSL=false&amp;allowPublicKeyRetrieval=true<
/property>
    <property name="connection.username">root</property>
    <property name="connection.password">1234</property>

    <property name="hibernate.dialect">org.hibernate.dialect.MySQL8Dialect</property>
    <property name="hibernate.show_sql">true</property>
```

240031280

```
        <property name="hibernate.format_sql">true</property>
        <property name="hibernate.hbm2ddl.auto">update</property>

        <mapping class="com.demo.Product"/>
    </session-factory>
</hibernate-configuration>
```

Build

From skill-3:

```
mvn -q compile
```

Run (with Maven [INFO] logs)

From skill-3:

```
mvn "-Dexec.mainClass=com.demo.ProductHqlDemo" org.codehaus.mojo:exec-maven-plugin:3.5.0:java
```

Expected console flow

- Creates/updates product table, wipes rows, inserts 6 sample products.
- Prints:
  - First 3 products (ids 1–3)
  - Next 3 products (ids 4–6)
  - Total and available counts
  - Group by description (Electronics/Furniture/Stationery)
- Ends with [INFO] BUILD SUCCESS (benign MySQL cleanup-thread warning may appear).

DB inspection (MySQL CLI)

• Created a **Maven project using Java 21**.
• Added **Hibernate ORM, Jakarta Persistence API, and MySQL Connector dependencies**.
• Implemented a **Student entity mapped to the student table** using the **IDENTITY primary key strategy**.
• Developed **StudentHqlDemo.java** which seeds sample student data and executes various **HQL queries**.
• Demonstrated HQL features including:

- Pagination using setFirstResult() and setMaxResults()
- Counting records using count()
- Grouping results using GROUP BY department
- Aggregate functions such as MIN() and MAX() on student marks
- Filtering students using BETWEEN condition on marks

• Configured **Hibernate with MySQL database fsad_skill3** using:

hibernate.show_sql = true
hibernate.format_sql = true
hibernate.hbm2ddl.auto = update

OUTPUTS:

24003128

New folder - skill-3-hql/src/main/java/com/klef/StudentHqlDemo.java - Spring Tools for Eclipse

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer | Project Explorer

- skill-3-hql
  - src/main/java
    - com.klef
      - Student.java
      - StudentHqlDemo.java
  - src/main/resources
  - src/test/resources
  - JRE System Library [JavaSE-21]
  - Maven Dependencies
  - target/generated-sources/annotations
  - target/generated-test-sources/test-annotations
  - src
  - target
  - pom.xml

Boot Dashboard

Type tags, projects, or working set names to match (incl. * and ? wildc

local

skill-3-hql/pom.xml | Student.java | StudentHqlDemo.java | hibernate.cfg.xml

```java
package com.klef;

import java.util.List;
import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.Transaction;
import org.hibernate.cfg.Configuration;

public class StudentHqlDemo {

private static final SessionFactory factory=new Configuration()
.configure("hibernate.cfg.xml")
.addAnnotatedClass(Student.class)
.buildSessionFactory();

public static void main(String[] args){

seedData();
runQueries();

factory.close();
}

private static void seedData(){

List<Student> students=List.of(
new Student("Vinay","CSE",85.5,20),
new Student("Rahul","ECE",78.0,21),
new Student("Ankit","AI",90.2,22),
new Student("Divya","CSE",88.0,21),
new Student("Kiran","ECE",72.5,23),
new Student("Sneha","AI",95.0,20)
);

```

Problems | Javadoc | Declaration | Console

```
<terminated> StudentHqlDemo [Java Application] C:\spring-tools-for-eclipse-5.0.1.RELEASE-e4.38.0-win32.win32.x86_64\sts-5.0.1.RELEASE\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win
    values
        (?, ?, ?, ?)
Students Inserted
```

Writable | Smart Insert | 102 : 2 : 2406

Search | ENG IN | 00:17 12-03-2026

New folder - skill-3-hql/src/main/java/com/klef/Student.java - Spring Tools for Eclipse

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer | Project Explorer

- skill-3-hql
  - src/main/java
    - com.klef
      - Student.java
      - StudentHqlDemo.java
  - src/main/resources
  - src/test/resources
  - JRE System Library [JavaSE-21]
  - Maven Dependencies
  - target/generated-sources/annotations
  - target/generated-test-sources/test-annotations
  - src
  - target
  - pom.xml

Boot Dashboard

Type tags, projects, or working set names to match (incl. * and ? wildc

local

skill-3-hql/pom.xml | Student.java | StudentHqlDemo.java | hibernate.cfg.xml

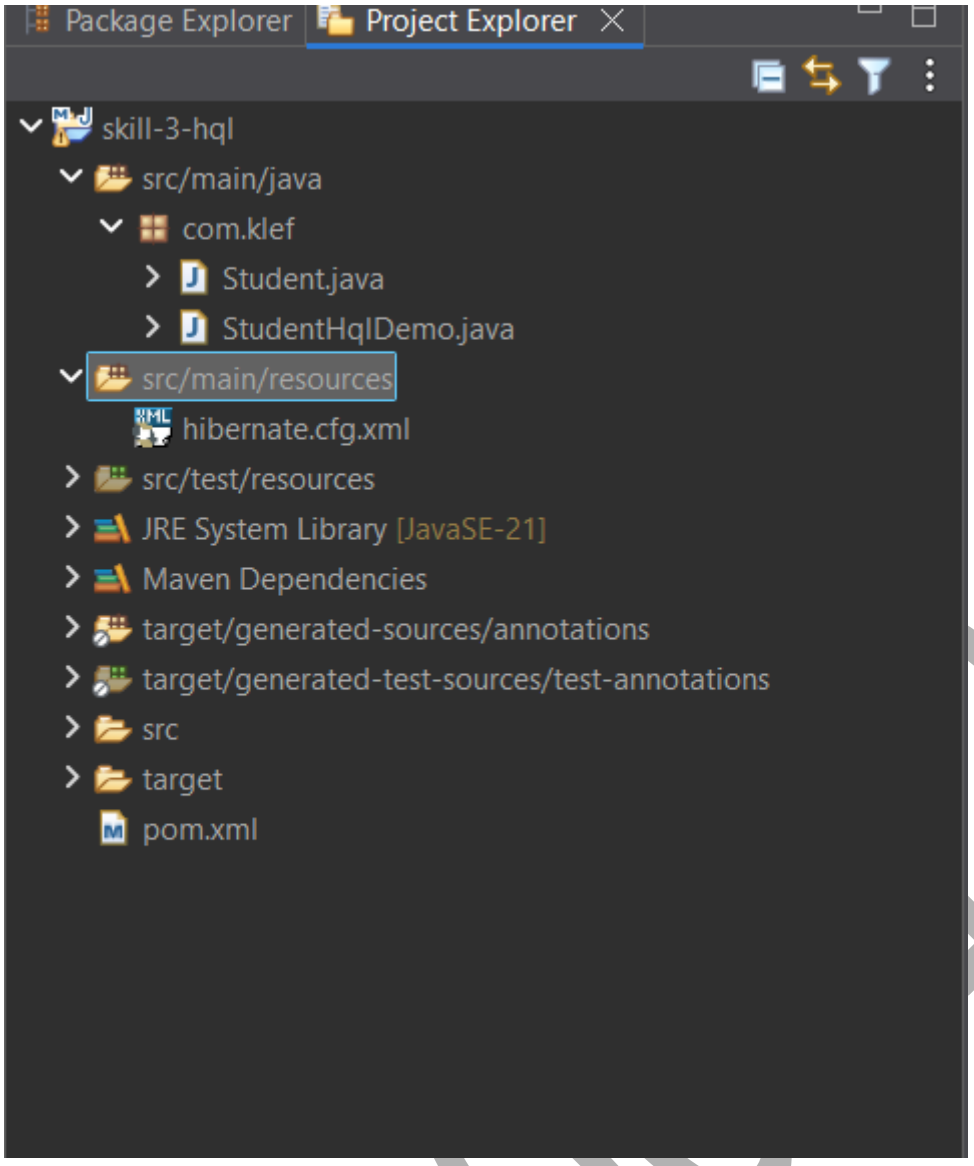
```java
package com.klef;

import jakarta.persistence.*;

@Entity
@Table(name="student")
public class Student {

@Id
@GeneratedValue(strategy = GenerationType.IDENTITY)
private Long id;

private String name;
private String department;
private Double marks;
private Integer age;

public Student(){}

public Student(String name,String department,Double marks,Integer age){
this.name=name;
this.department=department;
this.marks=marks;
this.age=age;
}

public Long getId(){ return id; }

public String getName(){ return name; }
public void setName(String name){ this.name=name; }

public String getDepartment(){ return department; }
public void setDepartment(String department){ this.department=department; }

```

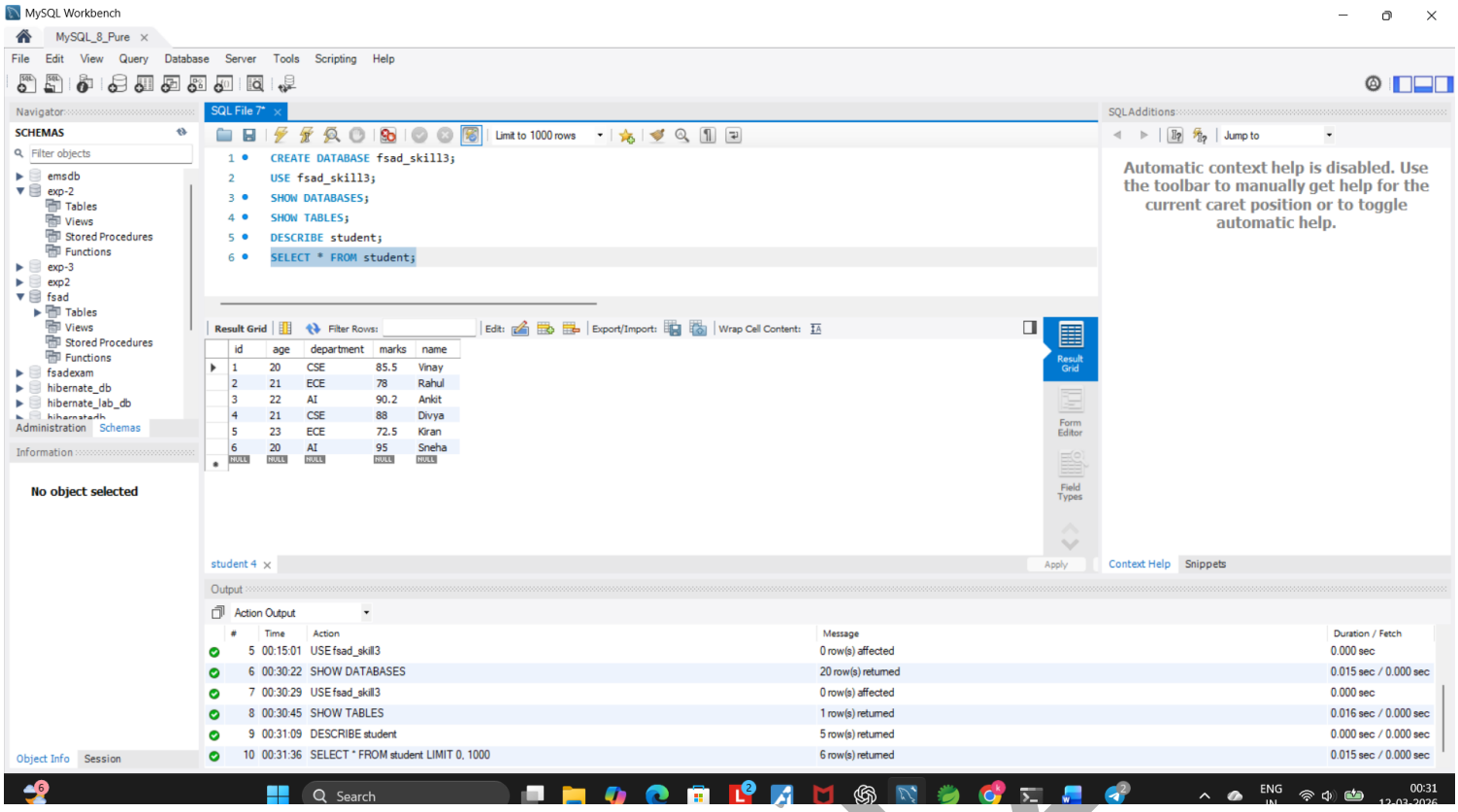
Problems | Javadoc | Declaration | Console

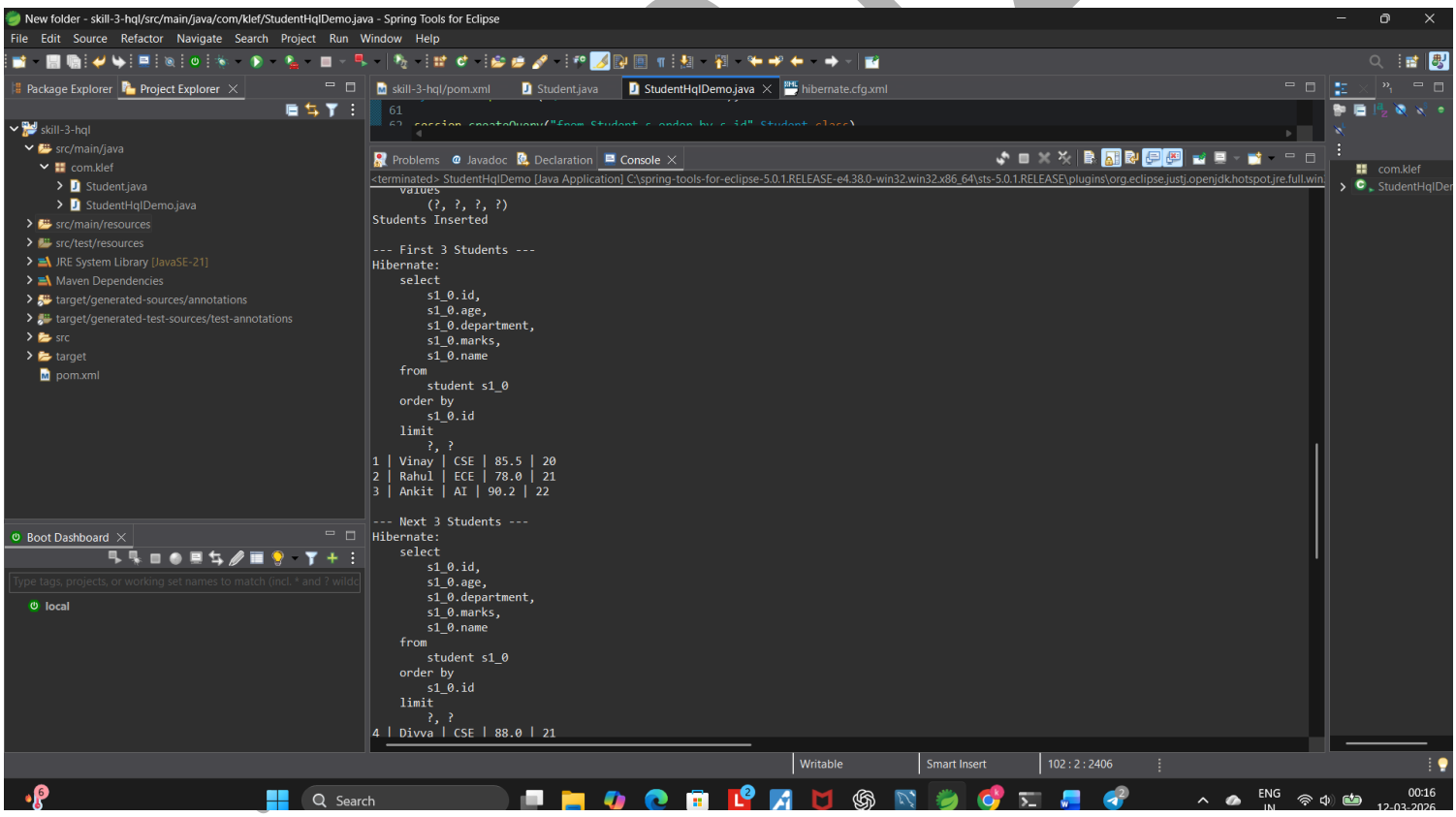
```
<terminated> StudentHqlDemo [Java Application] C:\spring-tools-for-eclipse-5.0.1.RELEASE-e4.38.0-win32.win32.x86_64\sts-5.0.1.RELEASE\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win
    values
        (?, ?, ?, ?)
Students Inserted
```

Writable | Smart Insert | 20 : 72 : 381

Search | ENG IN | 00:17 12-03-2026


Package Explorer
Project Explorer
skill-3-hql
src/main/java
com.klef
Student.java
StudentHqlDemo.java
src/main/resources
hibernate.cfg.xml
src/test/resources
JRE System Library [JavaSE-21]
Maven Dependencies
target/generated-sources/annotations
target/generated-test-sources/test-annotations
src
target
pom.xml





GitHub Repository Link

The complete source code for this experiment is available in the following GitHub repository:

Repository:
https://github.com/Kishtipati-Vinay-Bhargav-Reddy/2400031280_FSAD_SKILL_3